

Campus Services Portal - Project Documentation Index

This directory contains the formal documentation deliverables required by the **Business Requirements Document (BRD v2.0 - Angular, Section 12 Deliverables)** for the **Campus Services Portal**.

Deliverables Catalog

1. 05-Angular-ModuleCoverage-Matrix.md

- **Purpose:** Formal evidence mapping every Angular framework concept and curriculum requirement directly to active implementation file paths in the workspace.
- **Key Contents:**
 - Standalone Component Architecture & Bootstrap (`App.config.ts`)
 - Route Parameters, Query Parameters, Child/Nested Routes & Wildcard (**)
 - Functional Route Guards (AuthGuard, AdminGuard)
 - Lazy Loading (loadComponent code-splitting)
 - Navigation Progress Bar & Events
 - Root Singleton Services & `provideHttpClient()`
 - Reactive Forms, Dynamic FormArray, and Template-Driven Forms
 - Custom Validators (Index number, Campus email domain, Password match, Future dates)
 - Custom Directives (`AppHighlight`, `AppStatusColor`)
 - Custom & Built-in Angular Pipes (`shortText`, `statusLabel`, `imeRange`, `currency`, `date`)
 - Four Screen States Pattern (Loading, Error, Empty, Data)
 - Confirmation Dialogs & Destructive Action Interception
 - Angular Signals (`signal`, `computed`, `effect`) & Reactive State Management

2. Frontend-Documentation.md

- **Purpose:** Comprehensive technical inventory of the single-page application frontend.
- **Key Contents:**
 - **Component Inventory:** Exhaustive catalog of all **104 standalone components** with CSS selectors, file paths, inputs, outputs, and functional roles.
 - **Routing Table:** Complete map of all 20 routing modules, paths, route parameters, guards, and lazy-loaded components.
 - **Service Inventory:** Full register of all **27 typed Angular services**, API endpoints called, methods, and managed state.
 - **Forms Register:** Complete audit of Reactive Forms (dynamic controls, FormArrays, validation rules) and Template-driven Forms.

3. Swagger-OpenAPI-Documentation.md

- **Purpose:** Full RESTful API and OpenAPI / Swagger technical specification for the ASP.NET Core 8 backend.
- **Key Contents:**
 - Interactive Swagger UI Guide (<http://localhost:5000/swagger>) with JWT Bearer security locking.
 - Complete endpoint catalog across all **28 Controllers** and **60+ API endpoints**.
 - Detailed Request DTO schemas, Query/Route parameters, and HTTP response codes (200, 201, 400, 401, 403, 404, 409).

- Realistic example request and response JSON payloads for each module.
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Build & Quality Assurance Status

- **Frontend Angular Build:** `px ng build --configuration=development` → **0 Errors** (Application bundle verified)
- **Backend .NET Build:** `dotnet build --no-restore /t:Compile` → **0 Errors** (Build succeeded)